

Canadian Healthcare Interoperability: Existing Problem Report

Two-page project context report — the problem addressed by the Canadian Interoperable FHIR Health Server

Executive summary. Canada has extensive digital-health investment and many electronic record systems, but patient information often remains difficult to access, interpret, and use when a patient moves between care settings, organizations, or provincial boundaries. The fundamental problem is not simply that records are electronic or non-electronic. It is that records are held in systems with different identifiers, data models, terminology, access rules, and exchange capabilities. As a result, authorized clinicians may receive incomplete static documents instead of real-time, structured clinical data. The proposed project addresses this gap through a secure, federated, standards-based patient-summary exchange.

The Existing Problem

Canadian healthcare is delivered through provincially and territorially administered systems, with care further divided among regional health authorities, hospitals, primary-care practices, pharmacies, laboratories, diagnostic-imaging providers, long-term-care organizations, and specialized programs. Each organization may use different digital products, local workflows, data conventions, and privacy processes. A patient can therefore have multiple records, identifiers, medication lists, laboratory repositories, and document stores even while receiving continuous care in Canada.

The practical consequence is that a clinician in one setting may not be able to obtain an accurate, structured view of information created in another. A physician treating a patient who previously received care in another province may depend on patient recall, calls to another facility, faxed records, scanned PDFs, discharge documents, or manually re-entered information. These mechanisms may be slow, incomplete, difficult to search, or clinically ambiguous. They cannot reliably support automated reconciliation, decision support, safe alerting, or efficient continuity of care.

Why Electronic Records Alone Do Not Solve It

Electronic records solve local documentation and workflow problems, but an electronic record inside one organization is not automatically interoperable with another system. Data can be digitized yet remain technically inaccessible, semantically inconsistent, legally difficult to disclose, or operationally unavailable. A JSON API is also not enough if systems disagree about the meaning of codes, the identity of the patient, the freshness of the record, or the conditions under which access is permitted.

Interoperability layer	Current failure	Effect on care
Technical	PDFs, faxes, legacy interfaces, proprietary formats, inconsistent APIs	Data cannot be reliably queried, searched, or incorporated into workflow
Semantic	Local codes and variable terminology for conditions, labs, medications, and immunizations	Receiving system may not correctly interpret clinical meaning
Identity	Provincial identifiers and demographics do not form a universal patient key	Risk of missed matches, duplicate records, or wrong-patient linkage
Governance	Different custodianship, consent, privacy, and emergency-access rules	Authorized care teams may face unclear or delayed access pathways
Operational	Uneven gateway availability, lack of shared SLAs, and manual processes	Delays, partial information, additional workload, and inconsistent service

Clinical and Operational Consequences

The absence of reliable interoperability can affect both patient safety and system efficiency. A clinician who cannot see a current allergy or medication history may have to make decisions with incomplete information. A specialist who cannot access a recent laboratory result or diagnostic report may repeat testing or delay care while documents are obtained. At a transition of care, discrepancies between hospital medication orders, pharmacy dispensing records, and patient-reported medication use may go unresolved. These are not guaranteed outcomes in every case, but they are predictable risks when information cannot move in a structured, timely, provenance-preserving way.

The burden also falls on clinicians and administrative staff. They may need to request records by phone or fax, scan documents, reconcile duplicate data, re-enter information, and determine whether an external result is current or relevant. For patients, fragmentation can mean repeatedly explaining their history, carrying documents, undergoing duplicate assessments, or experiencing uncertainty over who can access their information. For health organizations, the result can be reduced efficiency, data-quality issues, integration cost, and difficulty measuring coordinated-care outcomes.

The Interprovincial Challenge

Cross-provincial care makes the problem more visible. A patient may move permanently, travel, attend post-secondary education, receive specialized treatment, or require emergency care outside their home province. The patient's source information may be held by providers or repositories that are technically and administratively separate from the receiving care team. Even where organizations use modern electronic systems, there may be no common, production-ready process for discovering the remote patient, confirming a safe match, evaluating consent and purpose of use, retrieving a standardized summary, validating terminology, preserving source attribution, and displaying any missing or stale information safely.

A provincial health number should not be assumed to be a pan-Canadian universal identifier. Nor should a name and date of birth be treated as definitive proof that two records belong to the same person. Identity resolution must be governed, conservative, auditable, and capable of routing uncertain cases to human review. A wrong match can create both a privacy breach and a clinical safety event; a missed match can prevent access to urgently needed information.

What Information Needs to Move First

A national solution does not need to exchange every element of every historical chart on day one. The highest-value starting point is a structured patient summary that supports time-sensitive and transition-of-care workflows. This includes patient identity and identifiers, active health concerns, allergies and intolerances, medications, immunizations, relevant laboratory observations, diagnostic reports, care-provider or organization attribution, and the provenance/freshness information needed to interpret these facts safely.

Priority domain	Why it matters in an initial exchange
Allergies and intolerances	Supports safer prescribing and urgent treatment decisions
Medication information	Supports medication reconciliation and reduces reliance on incomplete recall
Conditions/health concerns	Provides clinical context for triage, treatment, and specialist referral
Laboratory observations and reports	Avoids avoidable repetition and supports interpretation of recent care
Diagnostic reports	Provides findings and context from prior imaging or diagnostic workups
Immunizations	Supports continuity of preventive and public-health-related care
Provenance and freshness	Allows clinicians to judge who supplied data, when it was effective, and whether it may be stale or partial

Why Standardization Matters

The project relies on a layered Canadian standards approach. HL7 FHIR R4 provides the technical resource and API model. The Canadian Core Data for Interoperability (CACDI) defines standardized core health-data elements and value sets intended to support meaningful information flow. CA Core+ provides pan-Canadian FHIR profiles, extensions, and terminology artifacts aligned with the health-data-content work. PS-CA defines the structured patient-summary exchange context. Together, these materials provide a stronger foundation than generic FHIR CRUD endpoints alone.

Standardization is not only about choosing a format. It requires that systems agree on resource profiles, required elements, coding systems, value sets, profile versions, error behaviour, source attribution, and validation processes. A record that looks like FHIR

but lacks required clinical content, uses unverified local codes, or loses its source context may be technically transferable but not clinically dependable.

What This Project Is Designed to Solve

The Canadian Interoperable FHIR Health Server is designed as a prototype of a trusted federated exchange pattern. It will not replace provincial systems or create an unconditional central “single source of truth.” Instead, it will demonstrate how authorized systems can retrieve a priority patient summary while preserving the authority of source systems. The platform will use FHIR R4 resources and Canadian implementation artifacts; maintain provenance and freshness; support safe multi-identifier patient matching; enforce consent-aware, purpose-based authorization; validate terminology and profiles; and make partial or unavailable source data explicit.

Problem in current state	Project response
Static, unsearchable, or manually exchanged information	Structured FHIR resources, search, patient-summary Bundles, and machine-readable exchange
Inconsistent clinical meanings	Pinned profiles, terminology validation, source-code preservation, and controlled mappings
Uncertain patient matching	Multiple identifiers, deterministic/candidate matching, review queue, and no low-confidence auto-merge
Token-only access control	SMART/OIDC plus organization, role, purpose, patient context, consent, jurisdiction, and emergency policy checks
Loss of source/freshness context	FHIR Provenance, audit records, source metadata, timestamps, and data-state indicators
Remote system failures hidden from users	Gateway health, timeout/circuit breaker controls, OperationOutcome, and explicit partial-response display

Conclusion

The existing Canadian interoperability problem is a gap between the location of health information and the point at which it is needed for care. Addressing it requires more than a central database, a new EMR, or a generic API. It requires a federated design that connects technical exchange, common data content, terminology, patient identity, consent and privacy controls, provenance, clinical safety, operational resilience, and clinician workflow. This project begins with a deliberately narrow, testable use case—safe retrieval of a structured patient summary across simulated Ontario and British Columbia gateways—and establishes the architecture and controls needed to grow responsibly toward real-world connected care.

Key references: HL7 FHIR R4; Canada Health Infoway CA Core+ and PS-CA implementation materials; CIHI Canadian Core Data for Interoperability (CACDI) and Connected Care resources. Verify exact implementation-guide and terminology versions before any implementation or pilot.